

 MINI PROJECT

VEHICLE INSURANCE

EXPLORATORY DATA ANALYSIS



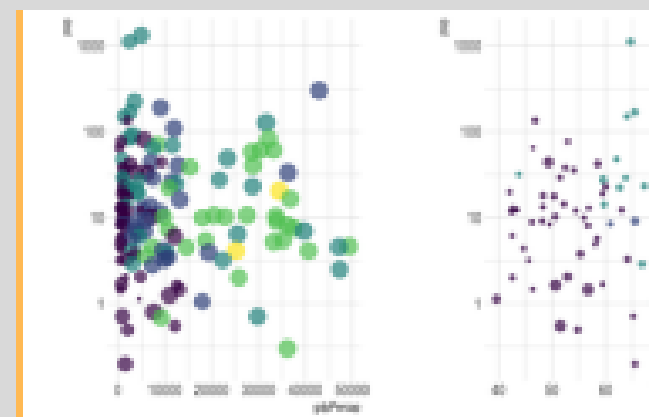
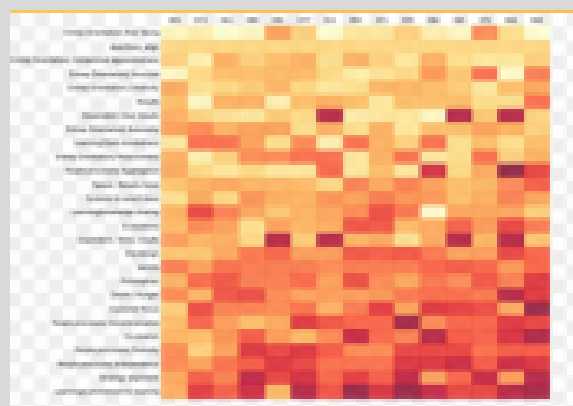
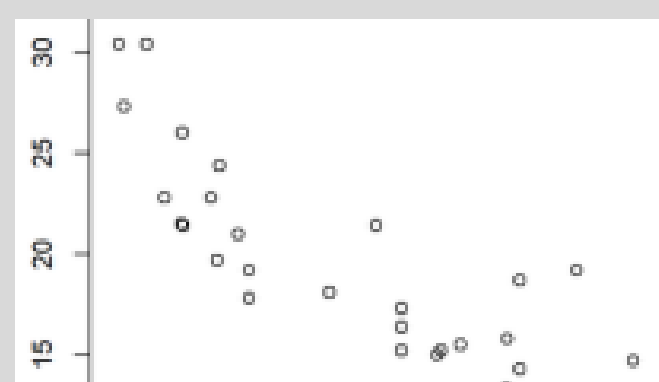
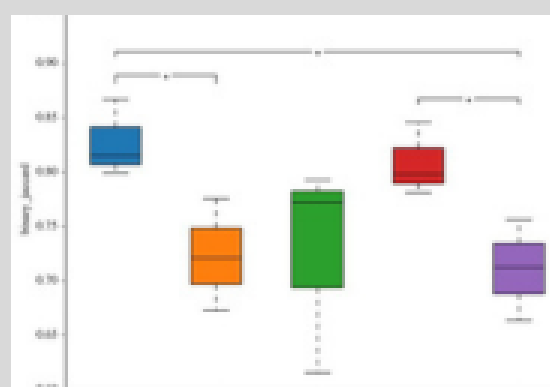
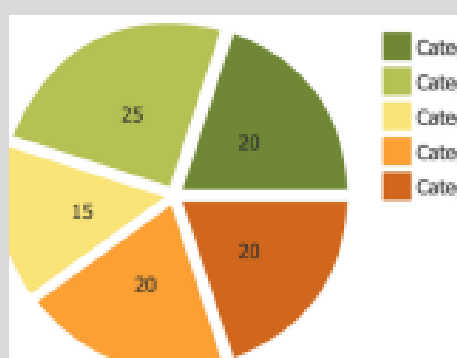
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| TOOLS



| GRAPHS



ABOUT THE INSURANCE

DATASET OVERVIEW:

THIS PROJECT CONDUCTED A DETAILED EXPLORATORY DATA ANALYSIS (EDA) ON A LARGE DATASET CONTAINING OVER 381,000 RECORDS OF VEHICLE INSURANCE APPLICANTS. THE MAIN OBJECTIVE WAS TO IDENTIFY KEY BEHAVIORAL AND DEMOGRAPHIC PATTERNS THAT INFLUENCE INSURANCE CLAIM RESPONSES. THROUGH SYSTEMATIC DATA PREPROCESSING, OUTLIER HANDLING, AND VISUALIZATION, SEVERAL SIGNIFICANT TRENDS AND RELATIONSHIPS WERE UNCOVERED. THE ULTIMATE GOAL IS TO DERIVE MEANINGFUL INSIGHTS THAT CAN INFORM DECISION-MAKING PROCESSES WITHIN THE INSURANCE DOMAIN.

OBJECTIVE:

THE PRIMARY OBJECTIVE OF THIS PROJECT IS TO CONDUCT AN IN-DEPTH EXPLORATORY DATA ANALYSIS (EDA) ON A DATASET RELATED TO VEHICLE INSURANCE. THROUGH THIS ANALYSIS, STUDENTS WILL GAIN VALUABLE INSIGHTS INTO THE PATTERNS, TRENDS, AND FACTORS INFLUENCING INSURANCE CLAIMS. THE PROJECT ENCOMPASSES VARIOUS ASPECTS OF DATA PREPROCESSING, VISUALIZATION, AND STATISTICAL ANALYSIS.

DESCRIPTION

- Loading and Preparing vehicle insurance Dataset for Analysis in Google Colab Notebook

```
[ ] from google.colab import drive  
drive.mount('/content/drive')
```

Mounted at /content/drive

- In this analysis , we import Pandas for data manipulation ,Numpy for numerical operations, Matplotlib for creating visualization ,and Seaborn and plotly for enhanced statistical graphic.

```
import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt  
import seaborn as sns  
import plotly.express as px
```

- We import the Google drive module to mount ‘Google Drive’, enabling access to files and dataset in the for our analysis.

```
x = "/content/drive/MyDrive/Bigbasket/Copy of Vehicle_Insurance.csv"  
  
df = pd.read_csv(x)
```

DATA OVERVIEW

- **Dataframe:**

```
df.head()
```

	id	Gender	Age	Driving_License	Region_Code	Previously_Insured	Vehicle_Age	Vehicle_Damage	Annual_Premium	Policy_Sales_Channel	Vintage	Response
0	1	Male	44	1	28.0	0	> 2 Years	Yes	40454.0	26.0	217	1
1	2	Male	76	1	3.0	0	1-2 Year	No	33536.0	26.0	183	0
2	3	Male	47	1	28.0	0	> 2 Years	Yes	38294.0	26.0	27	1
3	4	Male	21	1	11.0	1	< 1 Year	No	28619.0	152.0	203	0
4	5	Female	29	1	41.0	1	< 1 Year	No	27496.0	152.0	39	0

- **The dataset contains 381109 rows and 12 columns**

```
df.shape
```

```
(381109, 12)
```

- **data information**

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 381109 entries, 0 to 381108
Data columns (total 12 columns):
 #   Column                Non-Null Count  Dtype  
---  -
 0   id                    381109 non-null  int64  
 1   Gender                381109 non-null  object  
 2   Age                   381109 non-null  int64  
 3   Driving_License       381109 non-null  int64  
 4   Region_Code           381109 non-null  float64 
 5   Previously_Insured    381109 non-null  int64  
 6   Vehicle_Age           381109 non-null  object  
 7   Vehicle_Damage        381109 non-null  object  
 8   Annual_Premium        381109 non-null  float64 
 9   Policy_Sales_Channel  381109 non-null  float64 
10   Vintage               381109 non-null  int64  
11   Response              381109 non-null  int64  
dtypes: float64(3), int64(6), object(3)
memory usage: 34.9+ MB
```

**the dataset contains
9 numerical and 3
categorical variables**

Description of columns

The dataset being analysed offers an extensive array of information about vehicle insurance, including various details about insured individuals, their vehicles, and the insurance claims associated with them. The following is a thorough overview of the dataset's main components and variables:

- **Age:** The age of the insured person, which reflects their life stage and possible risk profile.
- **Gender:** The gender of the insured person, which could impact insurance premiums and the frequency claims.
- **Driving License:** The status or type of driving license held by the insured individual, which may include factors such as the license 's validity, class, and any endorsements or restrictions.
- **Region Code:** The geographic location of the insured individual, reflecting regional differences in risk factors and claim rates.
- **Previously Insured:** This refers to whether the insured individual had prior vehicle insurance coverage before their current policy. It can indicate their previous insurance history and may influence risk assessment and policy terms.

Description of columns

- **Vehicle Age:** The age or model year of the insured vehicle, indicating its condition and the potential risk of accidents or damage.
- **Vehicle Damage:** Details about the vehicle's damage status, which can influence insurance premiums and the frequency of claims.
- **Annual Premium:** The total amount of money paid by the insured individual for their vehicle insurance policy over the course of a year.
- **Policy Sales Channel:** The method or platform through which the insurance policy was sold to the insured individual. This could include channels such as direct sales, brokers, online platforms, or agents.
- **Vintage:** Typically refers to a vehicle that is at least 20 to 30 years old, depending on the definition used by different organizations or regions.
- **Response:** In a general context, a response is an answer or reaction to a question, request, or stimulus.

Description

- Statistical summary of numerical columns

```
df.describe()
```

	Age	Driving_License	Region_Code	Previously_Insured	Annual_Premium	Policy_Sales_Channel	Vintage	Response
count	370789.000000	370789.000000	370789.000000	370789.000000	370789.000000	370789.000000	370789.000000	370789.000000
mean	38.670344	0.997942	26.437281	0.459666	29264.643973	112.542923	154.345442	0.121781
std	15.440166	0.045316	13.310391	0.498371	14743.056656	54.039616	83.670478	0.327033
min	20.000000	0.000000	0.000000	0.000000	2630.000000	1.000000	10.000000	0.000000
25%	25.000000	1.000000	15.000000	0.000000	24102.000000	30.000000	82.000000	0.000000
50%	36.000000	1.000000	28.000000	0.000000	31319.000000	151.000000	154.000000	0.000000
75%	49.000000	1.000000	36.000000	1.000000	38596.000000	152.000000	227.000000	0.000000
max	85.000000	1.000000	52.000000	1.000000	61892.000000	163.000000	299.000000	1.000000

Insights:

- Dataset has 381,109 total records and no missing values in numeric columns.
- Age distribution shows that most customers are middle-aged.
- Annual _ Premium has a wide range, so scaling or outlier handling may be needed.
- Region _ Code and Policy _ Sales _ Channel are likely categorical features represented numerically.
- Vintage is fairly spread — indicates different customer tenures

Data cleaning & pre-processing

- This dataset has no null or missing values and neither it contains any duplicated values

```
df.isnull().sum()
```

0

Gender

0

Age

0

Driving_License

0

Region_Code

0

Previously_Insured

0

Vehicle_Age

0

Vehicle_Damage

0

Annual_Premium

0

Policy_Sales_Channel

0

Vintage

0

Response

0

dtype: int64

```
df.duplicated().sum()
```

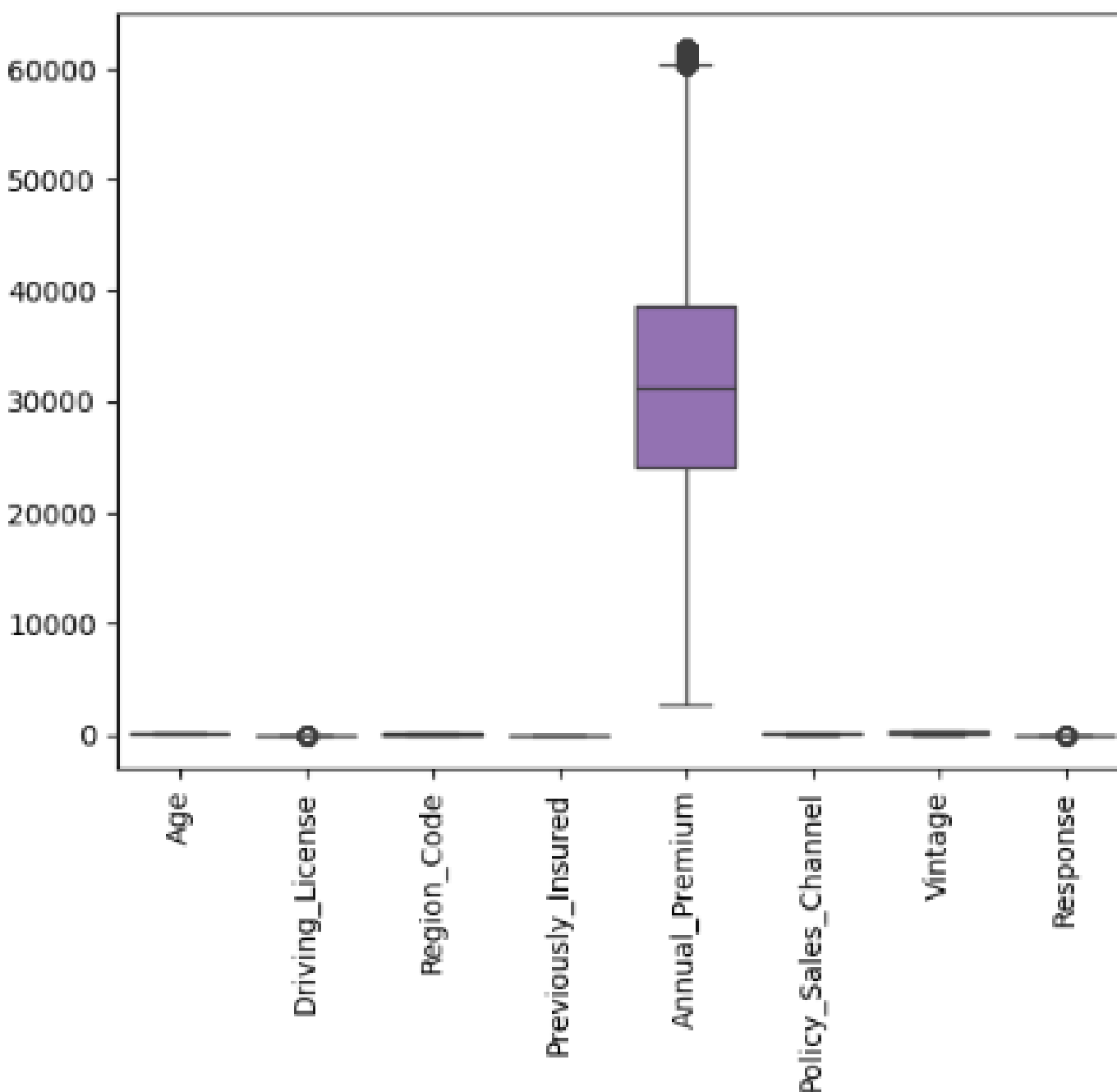
```
np.int64(0)
```



Data cleaning & pre-processing

- Now we check for outlier detection if there exist in any column.

```
sns.boxplot(df)  
plt.xticks(rotation=90)  
plt.show()
```



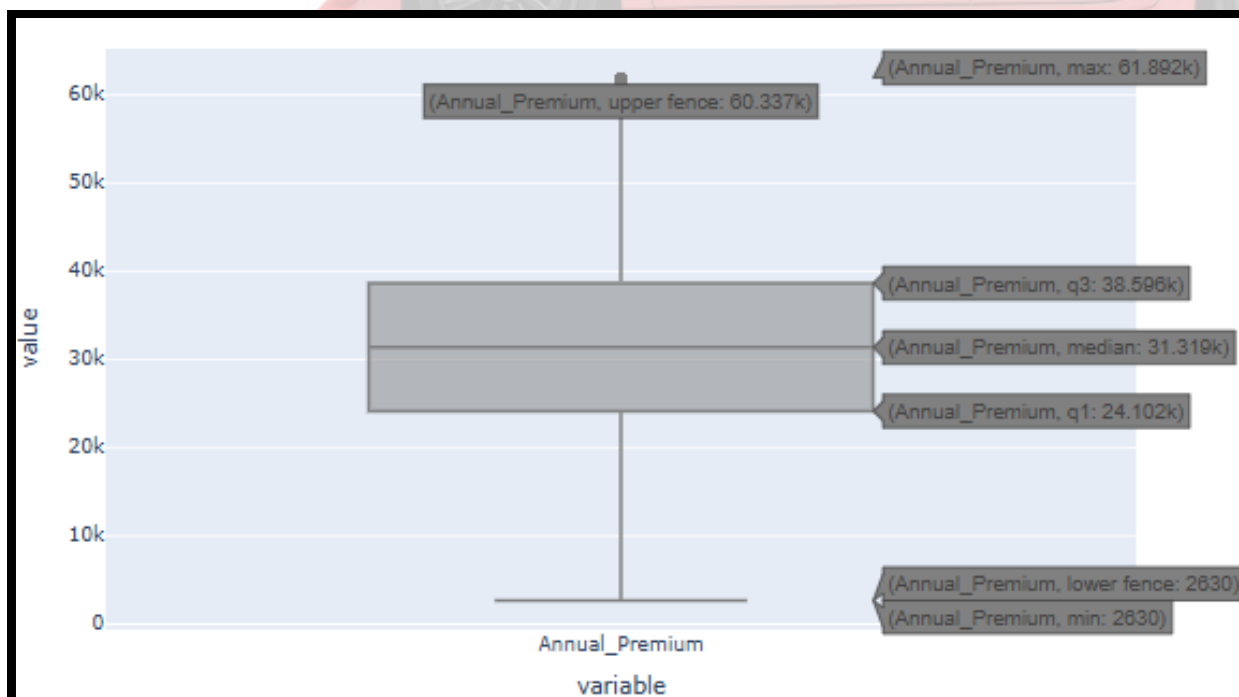
Data cleaning & pre-processing

- Annual premium columns is showing outliers and we have to fix this using the IQR method.

```
q1 = df['Annual_Premium'].quantile(0.25)
q3 = df['Annual_Premium'].quantile(0.75)
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr

df = df[(df['Annual_Premium'] < upper) & (df['Annual_Premium'] > lower)]
```

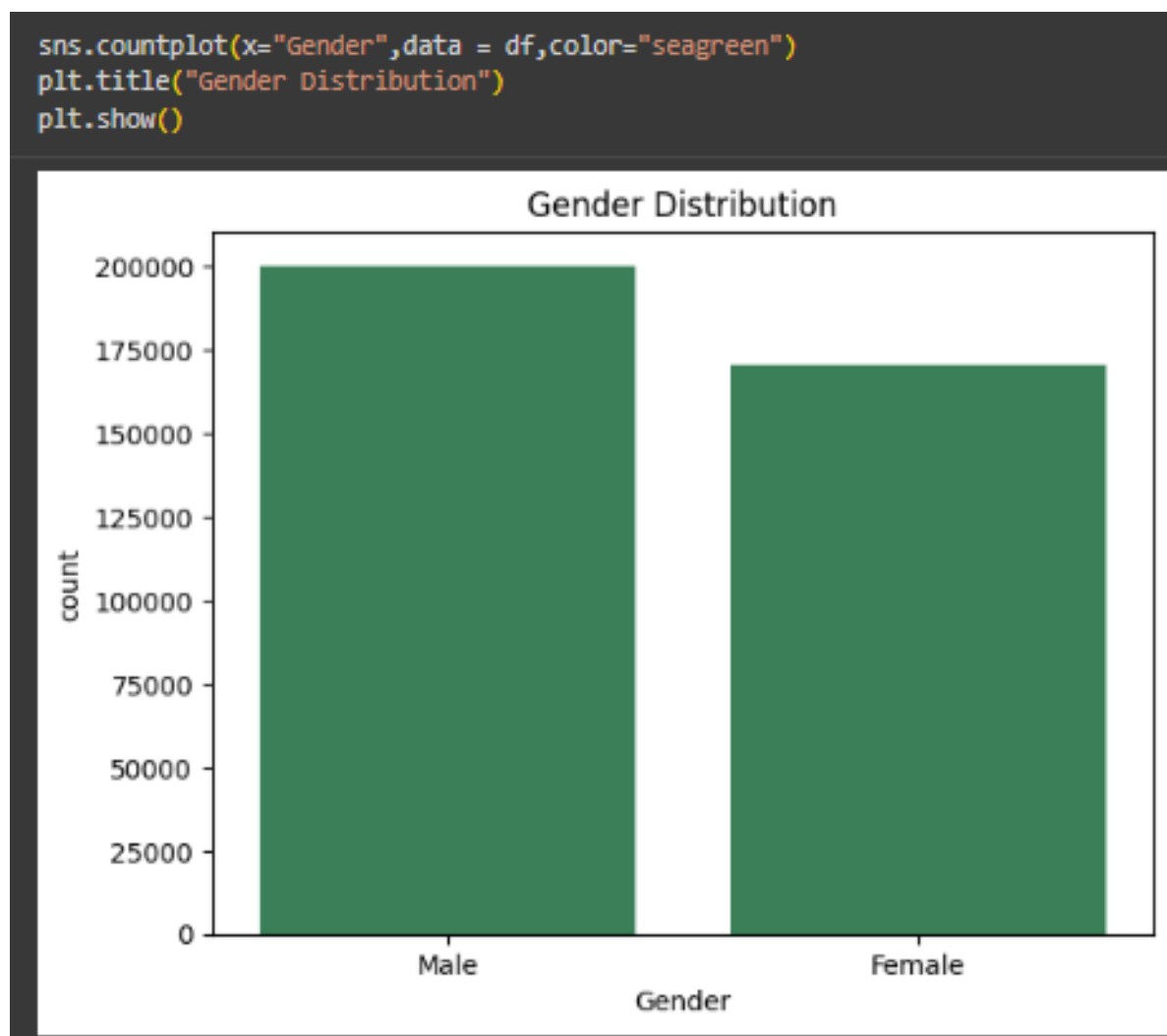
```
fig_2 = px.box(df['Annual_Premium'], color_discrete_sequence=['grey'])
fig_2.update_layout(width=800, height=500)
```



Insights: Initial premiums had extreme outliers above ₹100,000. After outlier removal, the majority are between ₹20,000 and ₹60,000. Most customers prefer mid-range premium plans. Outlier trimming improves model visualization clarity.

Data Visualization

- Gender Distribution



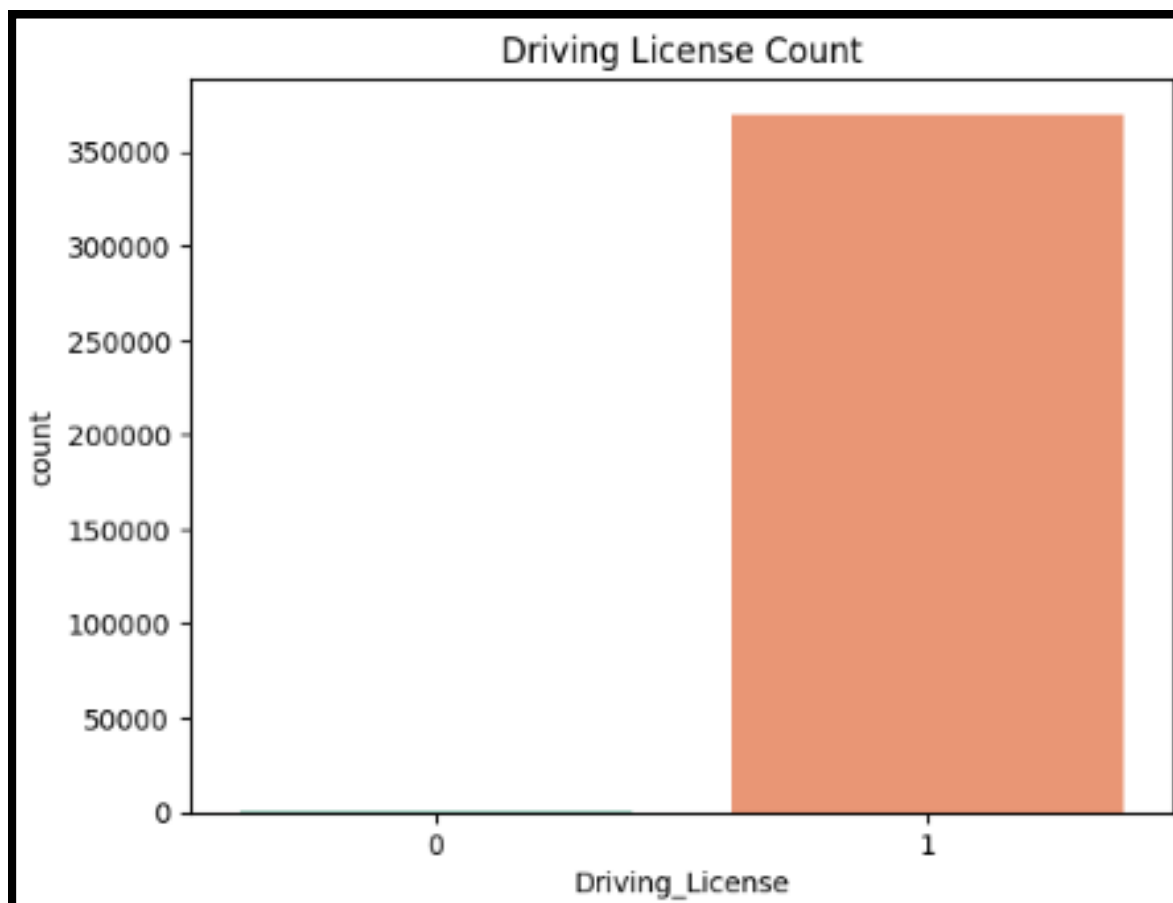
Insights: The gender distribution analysis shows a clear dominance of male customers in the vehicle insurance dataset. This suggests that men are more actively engaged in purchasing vehicle insurance policies. The company could explore marketing strategies aimed at increasing awareness and participation among female vehicle owners to achieve a more balanced customer base.

Targeted marketing can be designed to improve outreach to female customers.

Data Visualization

- Driving License

```
sns.countplot(x="Driving_License",palette="Set2",data=df)  
plt.title("Driving License Count")  
plt.show()
```



```
df['Driving_License'].value_counts()  
  
count  
  
Driving_License  
1      370026  
0         763
```

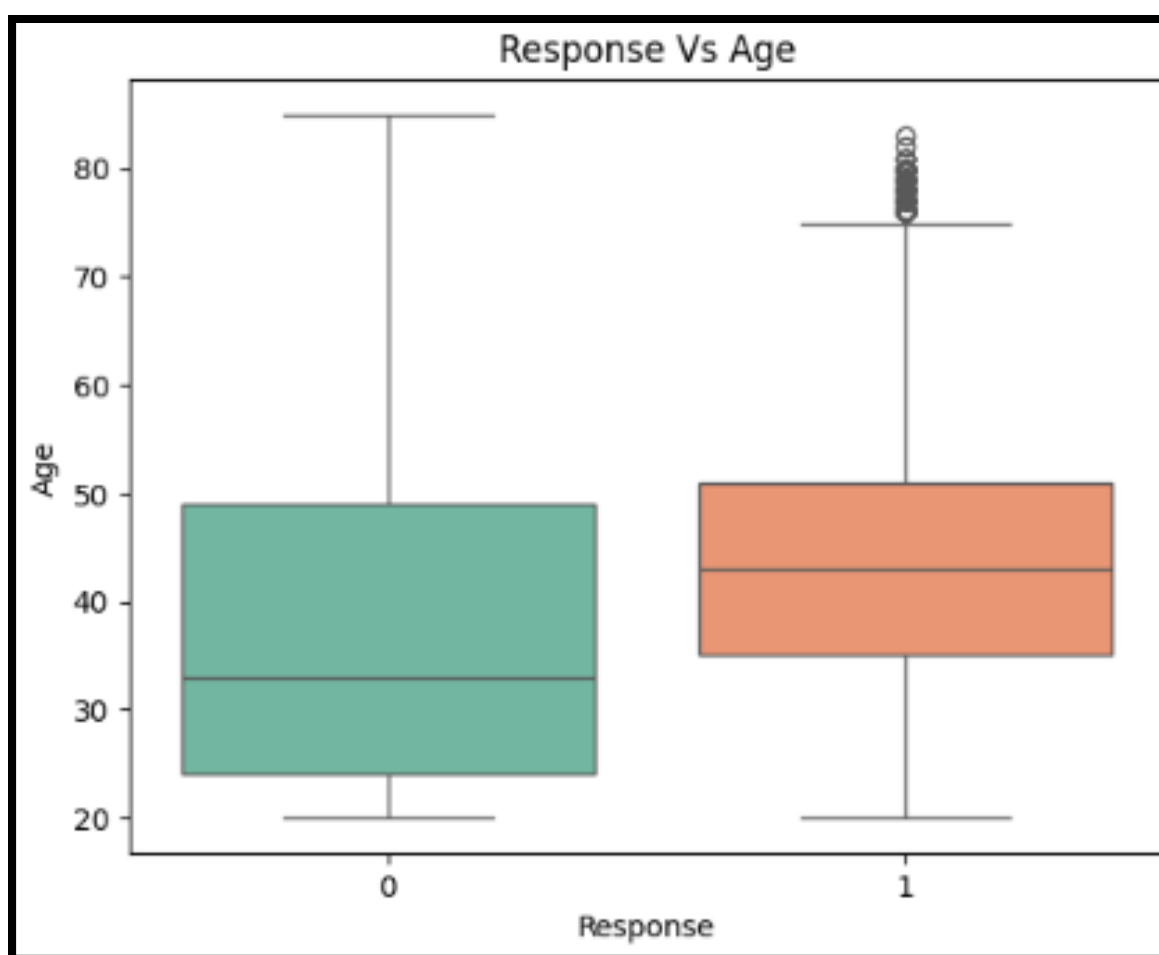
Insights:

The count plot illustrates the distribution of individuals based on their driving license status. It is observed that a large majority of the customers possess a valid driving license (Driving_License = 1), while only a small portion of the population does not hold one (Driving_License = 0). This suggests that most individuals in the dataset are eligible to drive, which may play a significant role in determining their likelihood of purchasing vehicle insurance. Having a driving license could indicate greater ownership or interest in vehicles, thereby influencing their insurance preferences.

Feature Analysis:

- Examine the relationship between features and the target variable (insurance claims).

```
sns.boxplot(x="Response",y="Age",data = df,palette='Set2')  
plt.title("Response Vs Age")  
plt.show()
```

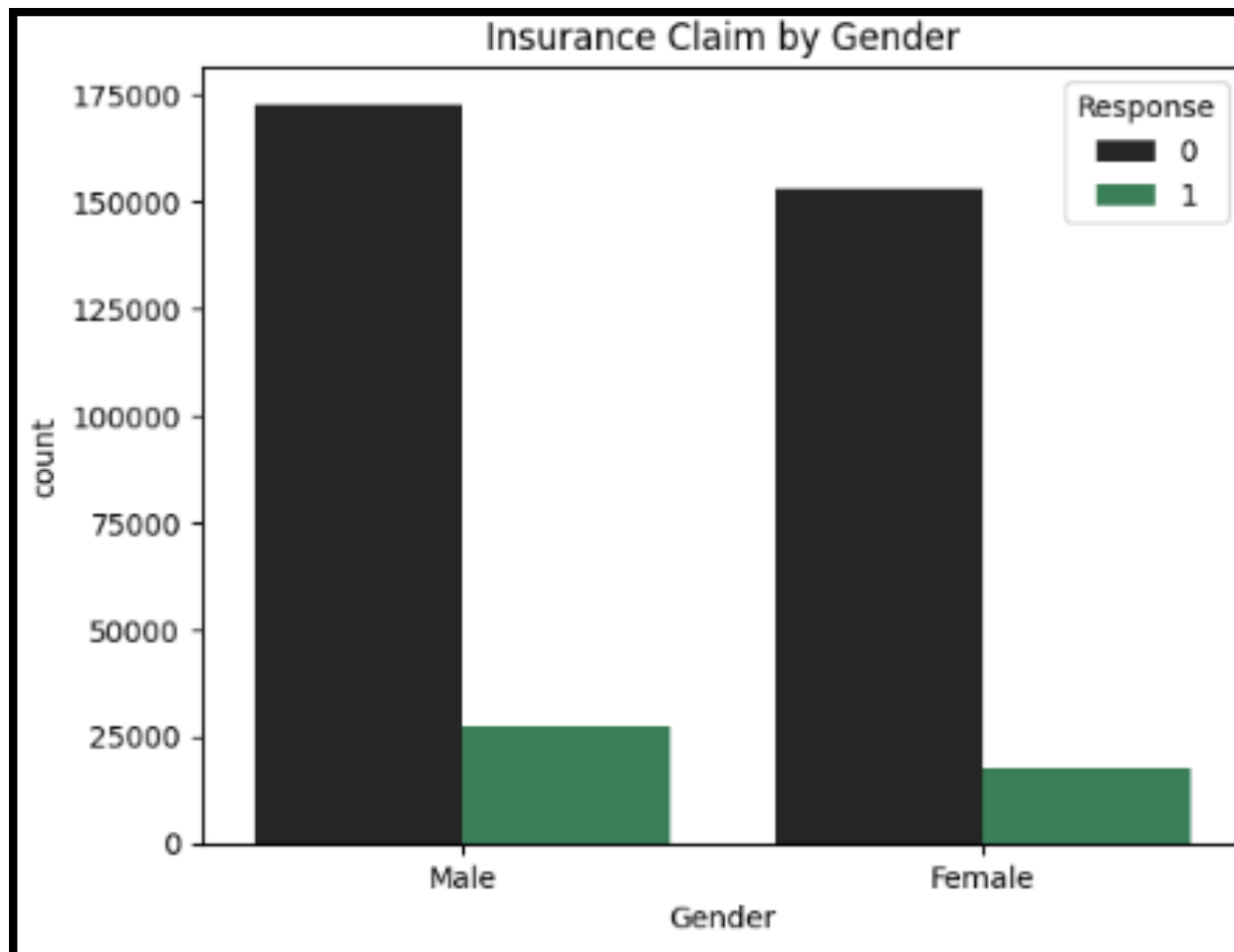


Insights: Most claims come from people aged between 25 and 50. This is the prime working-age population, possibly owning newer vehicles and having financial capability. Claims are rare from very young and very old age groups.

Feature Analysis

- Insurance claim by gender

```
sns.countplot(x="Gender",hue="Response",data = df,color="seagreen")  
plt.title("Insurance Claim by Gender")  
plt.show()
```

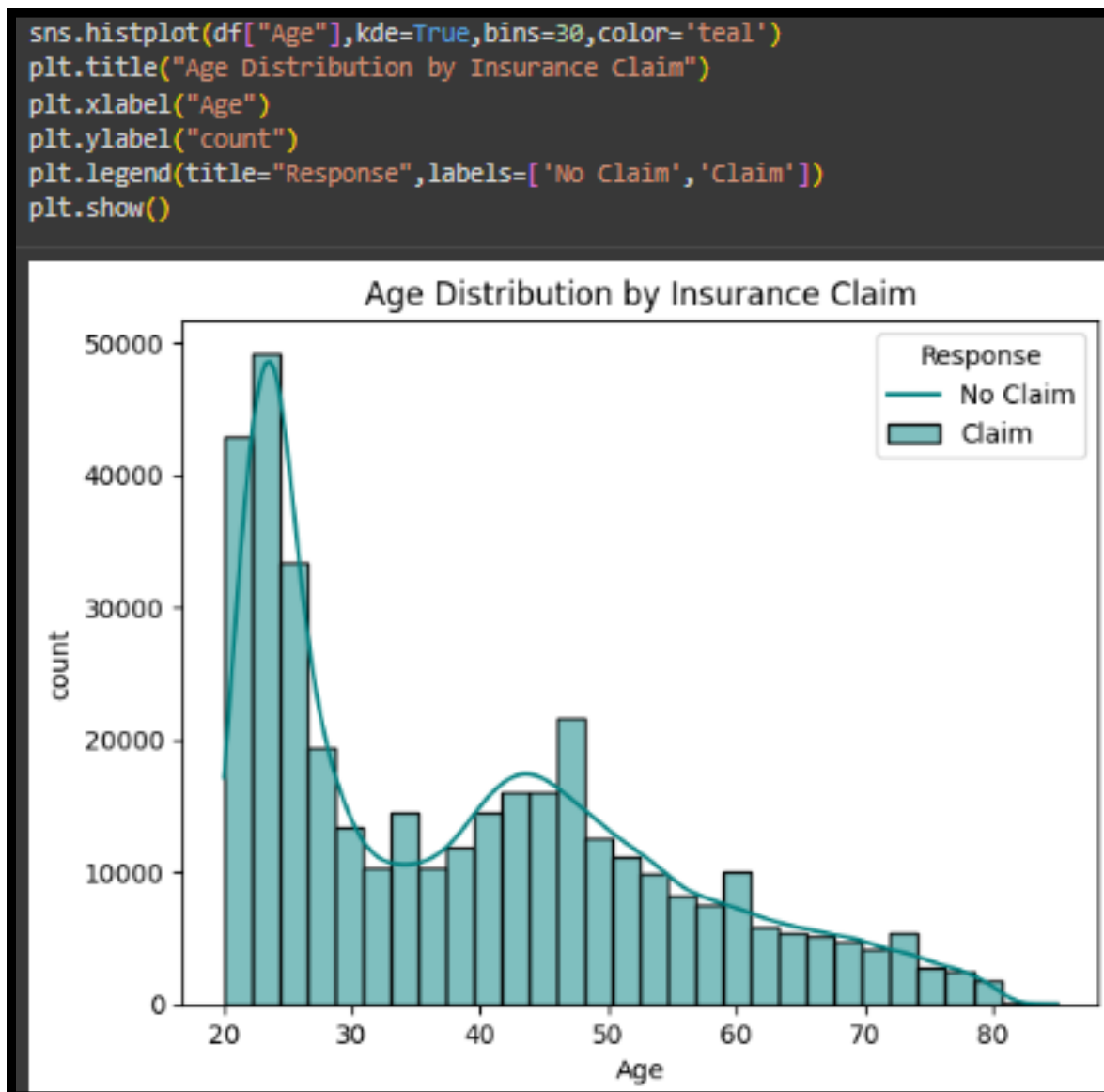


Insights:

The count plot represents the distribution of customers based on gender. It is observed that the dataset contains a higher number of male customers compared to female customers. This indicates that male individuals form the majority of the dataset, which may suggest that men are more likely to apply for or show interest in vehicle insurance. Understanding this distribution helps in analyzing how gender influences insurance preferences and response behavior.

Age Distribution

- Analyze the age distribution within the dataset and its impact on insurance claims.



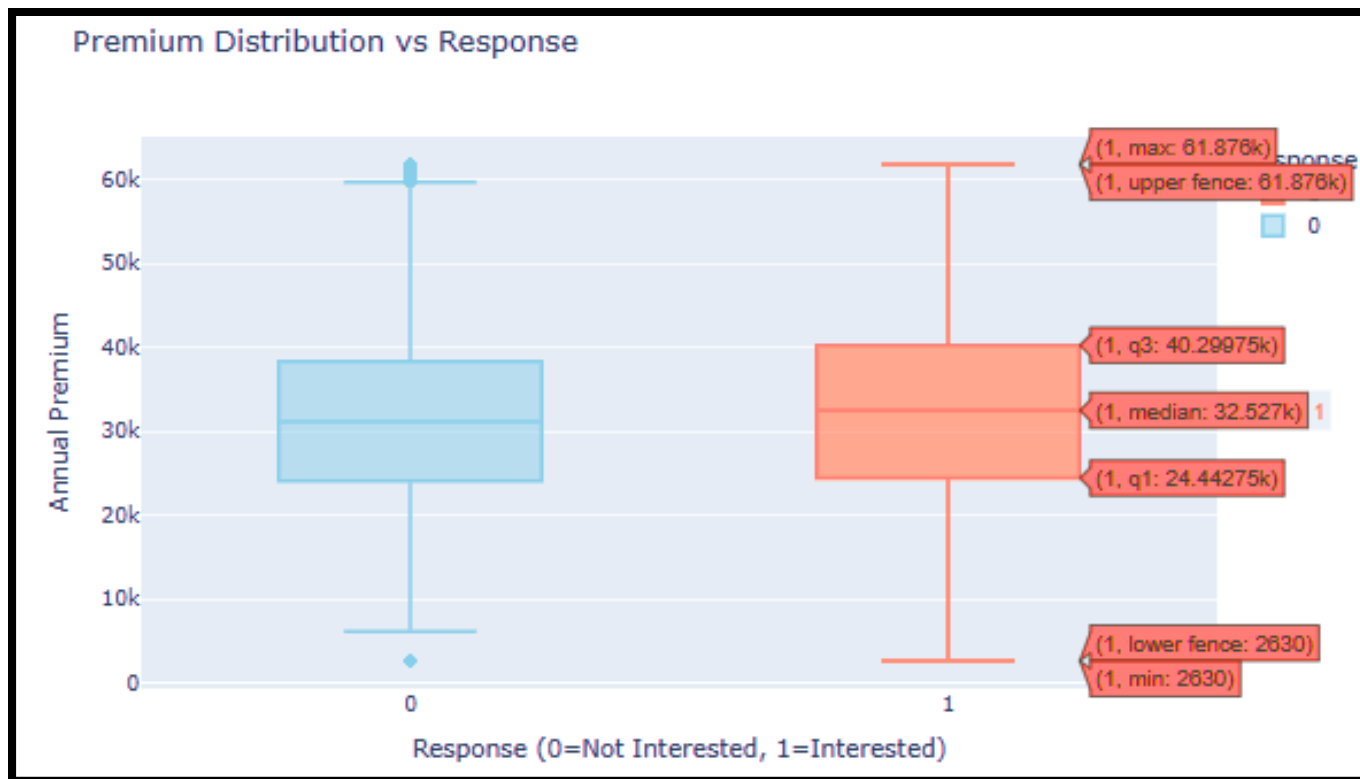
Insights:

Distribution is right-skewed with peaks in the 20s to 40s. Young and middle-aged individuals are the core market for vehicle insurance. Insurance companies should align policies and pricing for this demographic.

Premium Analysis:

- Investigate the distribution of insurance premiums and their correlation with claim frequencies.

```
fig = px.box(df,
             x='Response',
             y='Annual_Premium',
             color='Response',
             title='Premium Distribution vs Response',
             color_discrete_map={0:'skyblue',1:'orange'})
fig.update_layout(width=800, height=500,
                  xaxis_title="Response (0=Not Interested, 1=Interested)",
                  yaxis_title="Annual Premium")
fig.show()
```



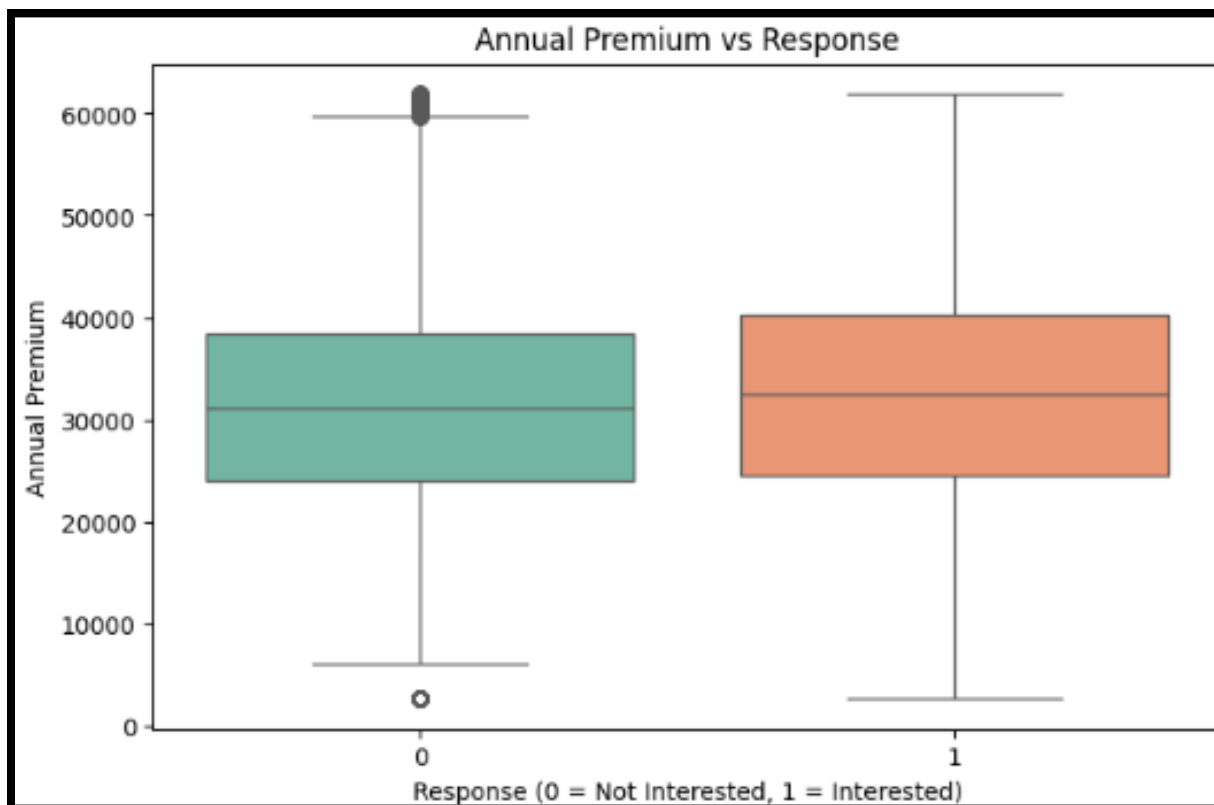
Insights:

The Premium vs Response analysis indicates that customers with moderate insurance premiums are more likely to show interest in policy renewal. High premium amounts appear to reduce customer interest, highlighting the importance of balanced and competitive pricing strategies. The presence of high-value outliers also suggests a niche segment that may benefit from personalized insurance plans.

Claim Frequency

- Investigate the distribution of insurance premiums and their correlation with claim frequencies.

```
plt.figure(figsize=(8,5))
sns.boxplot(x="Response", y="Annual_Premium", data=df, palette="Set2")
plt.title("Annual Premium vs Response")
plt.xlabel("Response (0 = Not Interested, 1 = Interested)")
plt.ylabel("Annual Premium")
plt.show()
```

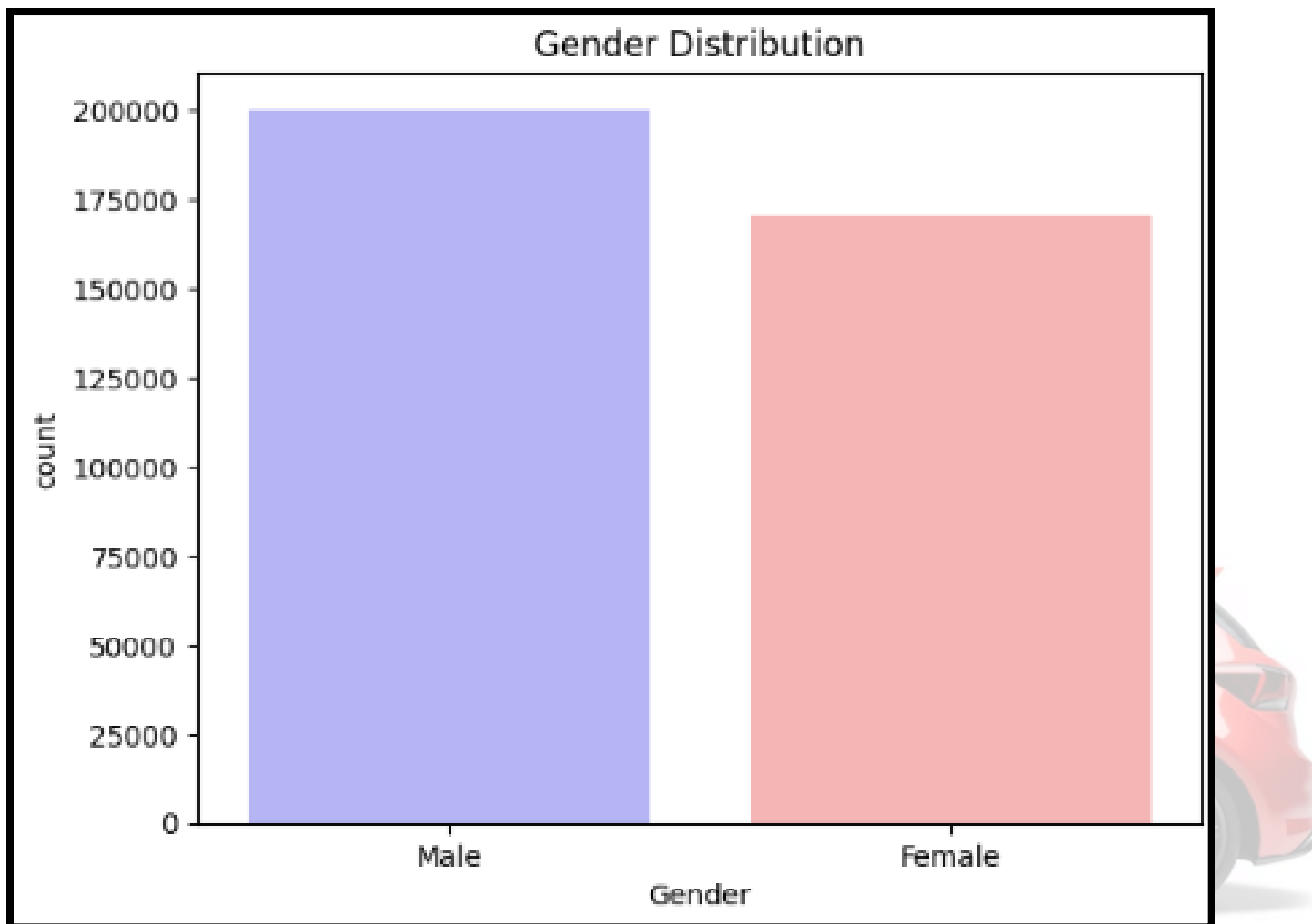


Insights:

The analysis explores the relationship between the annual premium amount and customer response toward purchasing insurance. The boxplot shows that the distribution of premiums for both interested (Response = 1) and non-interested (Response = 0) customers is quite similar, with no major difference in median values. This indicates that the premium amount does not have a strong influence on customer interest in buying vehicle insurance. Other factors such as age, vehicle age, or previous insurance history may play a more significant role in determining customer decisions.

Gender Analysis:

- Investigate the role of gender in insurance claims.

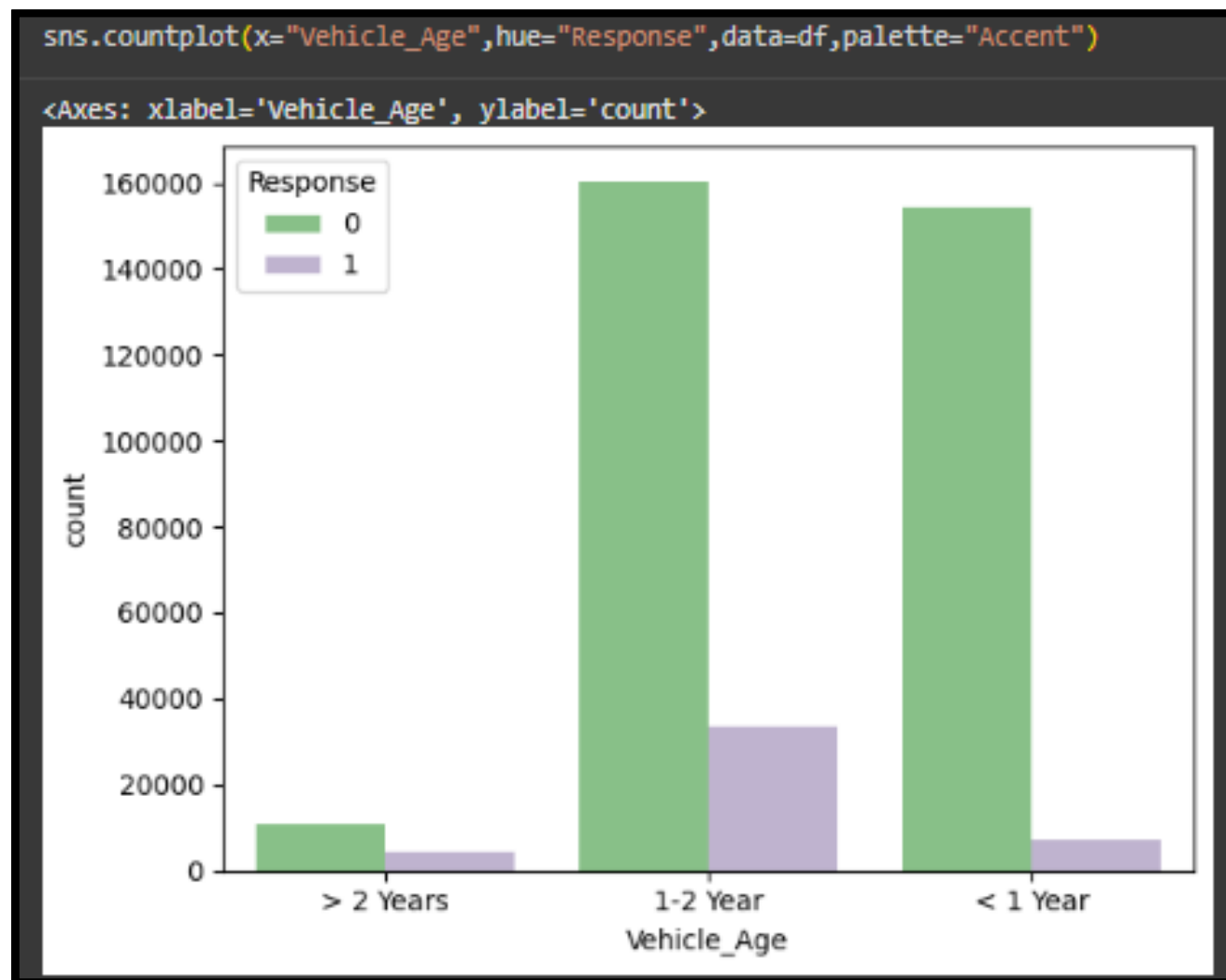


Insights:

The gender distribution analysis shows a clear dominance of male customers in the vehicle insurance dataset. This suggests that men are more actively engaged in purchasing vehicle insurance policies. The company could explore marketing strategies aimed at increasing awareness and participation among female vehicle owners to achieve a more balanced customer base. Targeted marketing can be designed to improve outreach to female customers.

Vehicle Age and Claims:

- Examine the impact of vehicle age on the likelihood of a claim



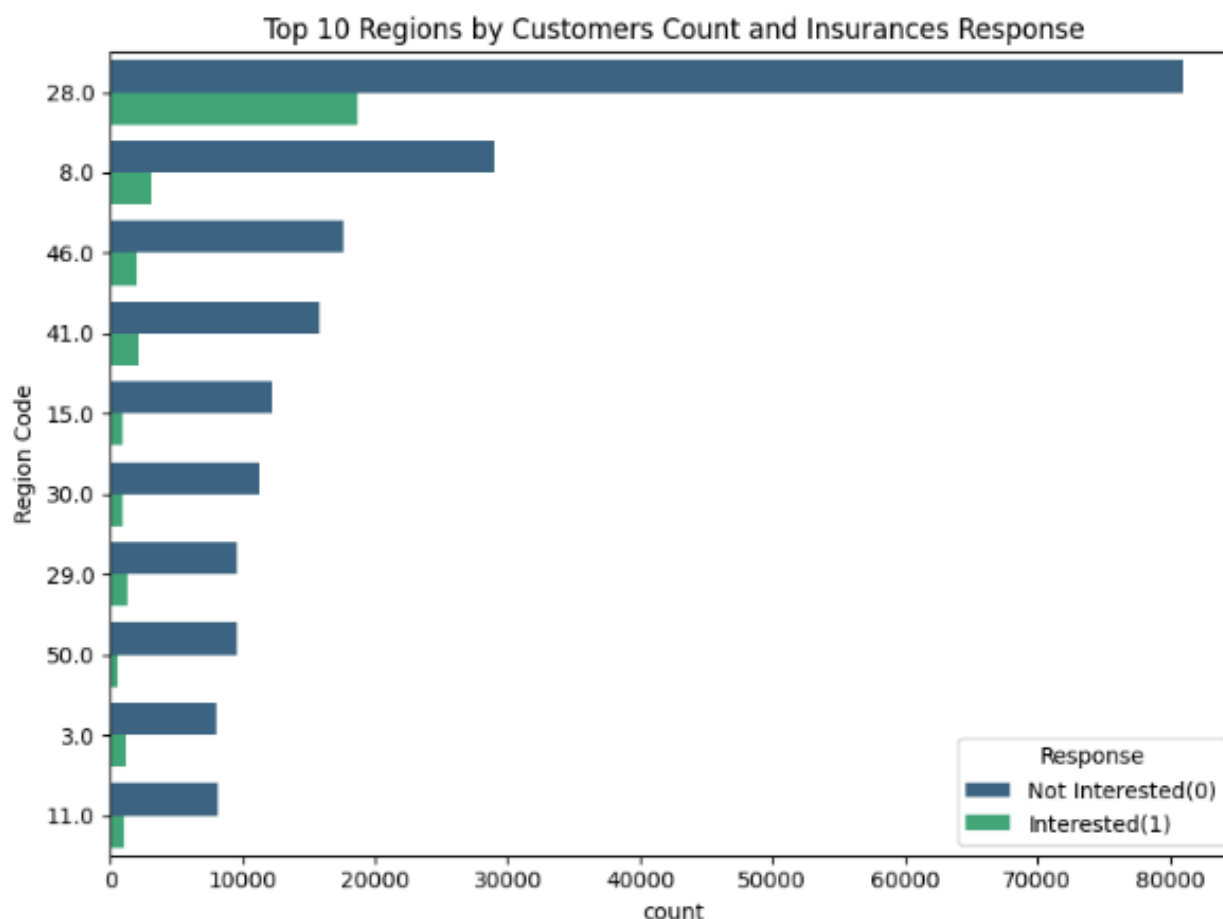
Insights:

Vehicles aged 1–2 years have the highest claim interest, followed by <1 year. Insight: Newer vehicles are more frequently insured and claimed. Older vehicles (>2 years) are less likely to be involved in claims, possibly due to depreciation or being uninsured.

Region-wise Analysis:

- Analyze regional patterns in insurance claims

```
top_regions= df["Region_Code"].value_counts().nlargest(10).index
df_top_regions = df[df["Region_Code"].isin(top_regions)]
plt.figure(figsize=(8,6))
sns.countplot(y="Region_Code", hue="Response",data=df_top_regions,
              order=top_regions,palette="viridis")
plt.title("Top 10 Regions by Customers Count and Insurances Response")
plt.ylabel("Region Code")
plt.legend(title="Response",labels=["Not Interested(0)","Interested(1)"])
plt.tight_layout()
plt.show()
```

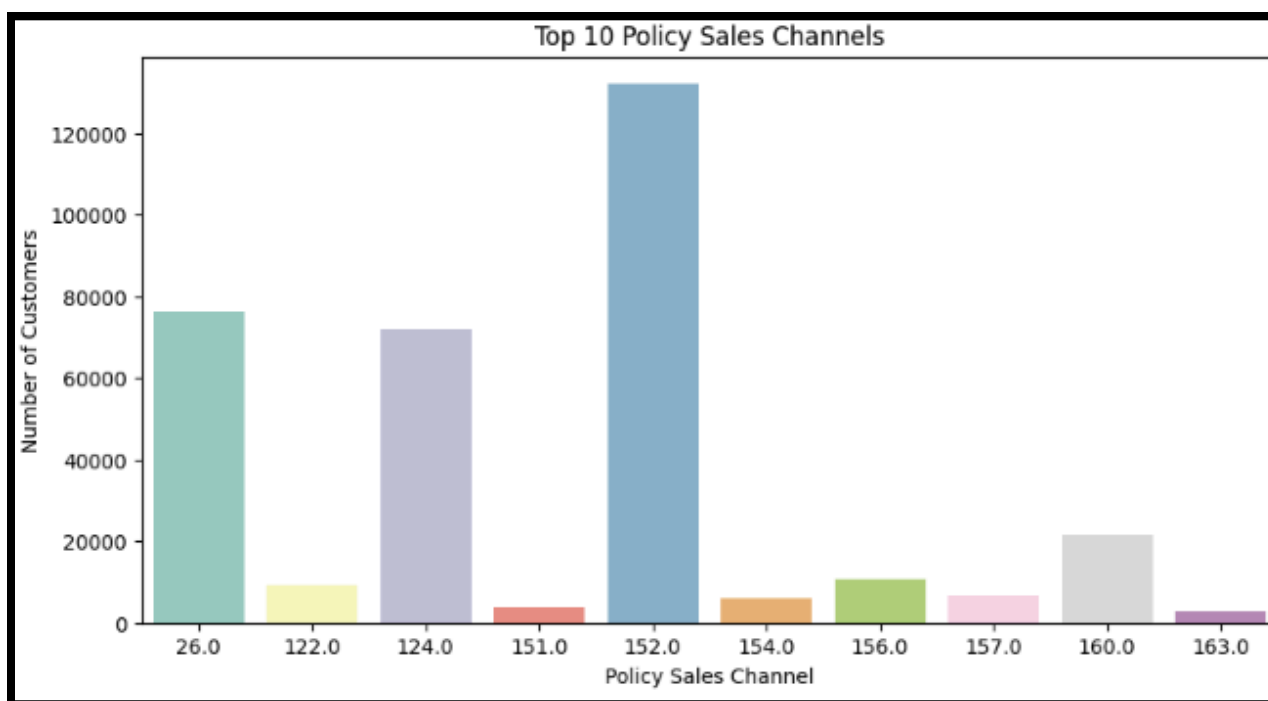


Insights: the regional analysis reveals that while a few key regions drive most of the customer engagement, others show lower interest levels — highlighting the need for region-specific marketing and customer retention strategies.

Policy Analysis

- Explore the distribution and impact of different insurance policy types.

```
plt.figure(figsize=(10,5))
top_channels = df["Policy_Sales_Channel"].value_counts().nlargest(10).index
sns.countplot(data=df[df["Policy_Sales_Channel"].isin(top_channels)],
              x="Policy_Sales_Channel", palette="Set3")
plt.title("Top 10 Policy Sales Channels")
plt.xlabel("Policy Sales Channel")
plt.ylabel("Number of Customers")
plt.show()
```



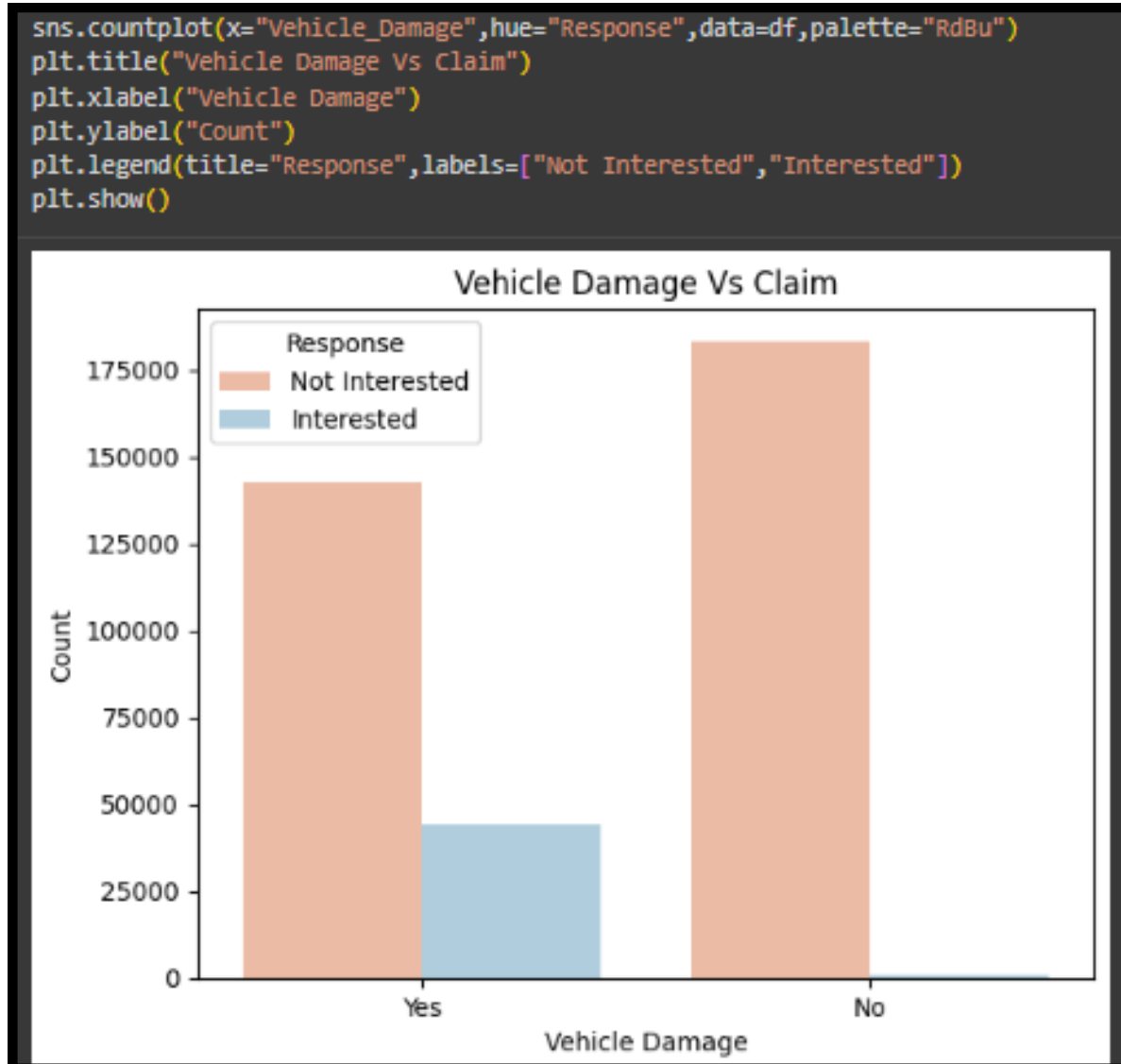
Insights:

The count plot illustrates the distribution of customers across various policy sales channels. It is observed that a few channels dominate the dataset, indicating that most insurance policies are sold through a limited number of highly active channels. This suggests that these channels are either more accessible to customers or more effective in reaching potential policyholders. Conversely, several other channels have relatively low participation, implying limited usage or lower customer reach.

Overall, the analysis highlights that a small number of sales channels contribute to the majority of policy sales, reflecting an uneven distribution in the way insurance policies are marketed and sold.

Claim Frequency by Vehicle Damage:

- Investigate the relationship between vehicle damage and claim frequencies.



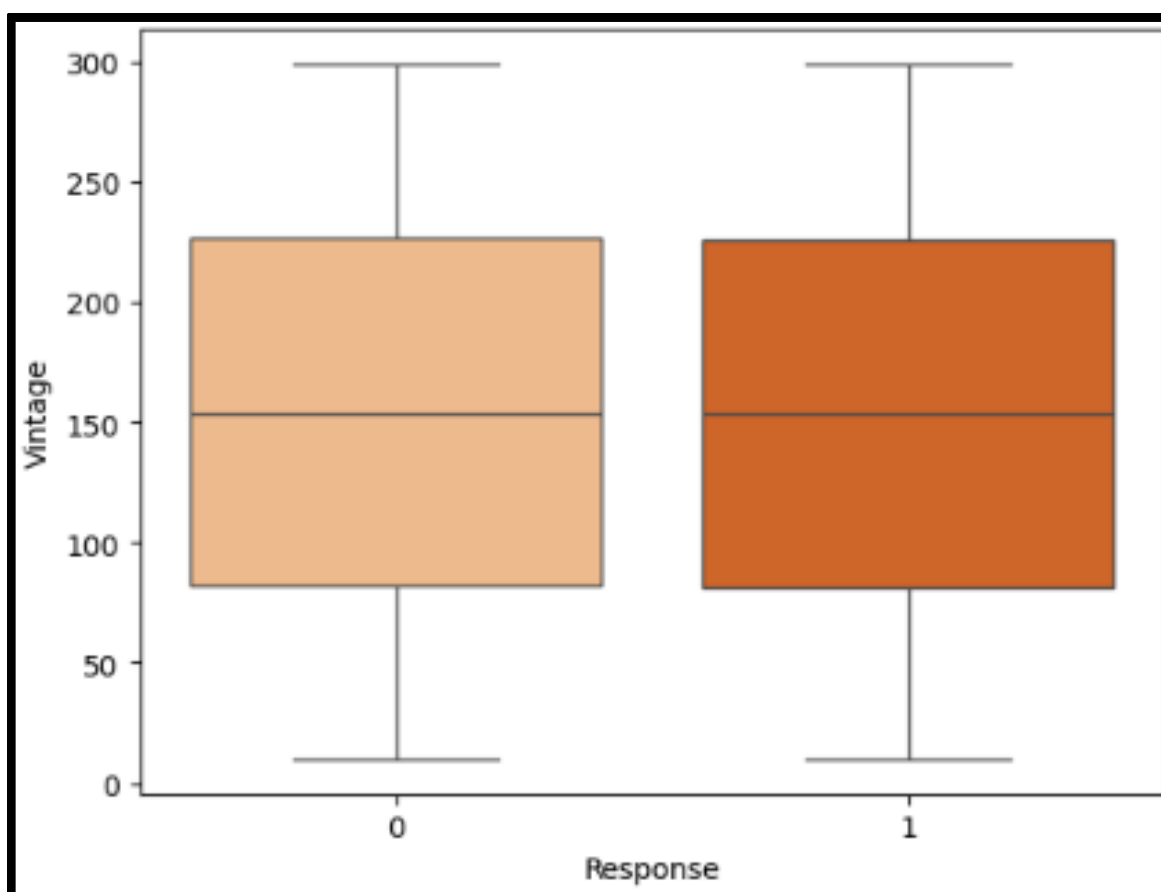
Insights:

- Users with vehicle damage ("Yes") show significantly higher response rates than those without.
- Past vehicle damage strongly influences the decision to purchase or claim insurance. It's a major risk indicator.

Customer Loyalty:

- Analyze if the number of policies held by a customer influences claim likelihood.

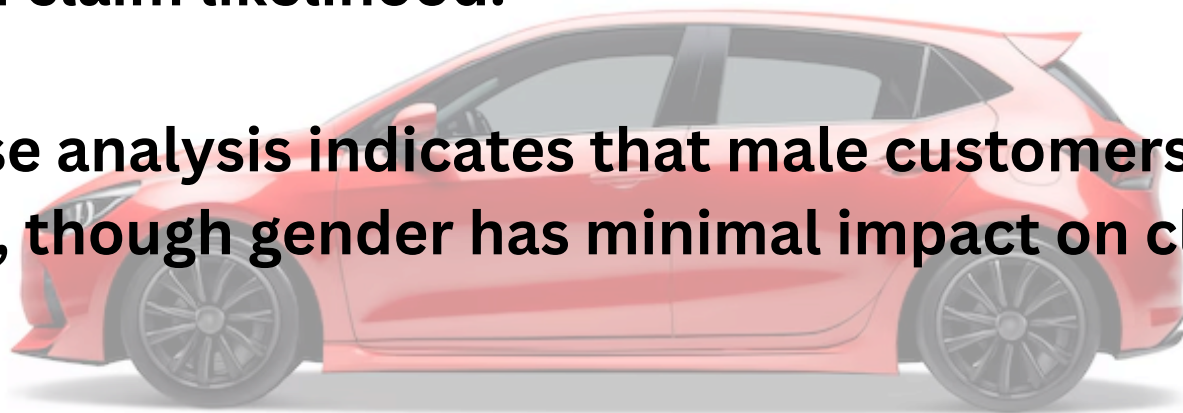
```
sns.boxplot(x="Response",y="Vintage",data=df,palette="Oranges")
```



Insight: Claim interest doesn't vary significantly with how long a customer has been associated. New or old customers may claim equally.

Final Report

- The exploratory data analysis (EDA) of the Vehicle Insurance dataset provides key insights into customer demographics, vehicle details, and insurance behavior.
- The age distribution reveals that most customers fall in the middle-age group, indicating a balanced risk segment.
- Premium analysis shows that the majority of policies have moderate premium values, with no strong link between premium amount and claim likelihood.
- Gender-wise analysis indicates that male customers dominate the dataset, though gender has minimal impact on claim response.
- Vehicle age and damage status significantly influence claim frequencies, as older and previously damaged vehicles show higher claim tendencies.
- Region-wise and policy channel analyses reveal uneven distribution, with a few regions and channels driving most sales. Lastly, customer loyalty analysis suggests that previously insured individuals are more likely to respond positively.
- Overall, the EDA highlights key behavioral patterns useful for risk assessment and strategic decision-making in vehicle insurance.



Thanks for reading

for coding part.....

